

Maximilian Müller

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Education

- 2021 – now **ELLIS PhD student in Machine Learning, University of Tübingen**
Focus on safe, robust and reliable Machine Learning, Generalization and Optimization. Supervisors: Matthias Hein and Gergely Neu.
- 2018 – 2021 **M.Sc. Physics (Ø 1, 2) and B.Sc. Physics, LMU Munich (Ø 1, 2)**
Master thesis at Max Planck Institute for Plasma Physics (Ø 1, 0):
Uncertainty quantification with Bayesian Neural Nets for Machine Learning - based fluid simulations
- 2019 – 2020 **M.Sc. Data Science, Barcelona GSE (Ø 9.6/10)**
Highest final grade of my cohort
Thesis title: *Scalable Inference for Crossed Random Effects Models (Ø 9.5/10)*.
- 2013 **High School - Abitur (Ø 1, 0)**

Practical Experience

- June 2025 - now **Research Intern, Google Deepmind, Zürich**
will work on Image Generation in GenAI team
- 2023 - now **Red Teamer, OpenAI, remote as indep. contractor**
Various red teaming efforts to assess the risks and safety profile of OpenAI models and systems like the Operator Agent, GPT-4 and GPT-4o as part of the OpenAI Red Teaming Network.
- 2020 – 2021 **Research Assistant in Bayesian Statistics, Bocconi University, remote**
Development of scalable algorithms for Bayesian statistics.
- 2019 **Junior IT Consultant TNG Technology Consulting, Munich**
Developed and evaluated investing strategies, set up end-to-end architecture for IoT-showcase and developed recommendation system for internal learning platform.
- 2018 **Working Student in Quantum Optics** Toptica Photonics AG, Munich
Developed experiment for doppler-free spectroscopy of iodine.
- Research Intern in Particle Physics** University of Alberta, Canada
Bubble chamber simulations with Geant4 (C++ based program) for dark matter detection.

Selected Publications

Conference and Workshop Proceedings

- 1 **M. Müller** and M. Hein, “Mahalanobis++: Improving ood detection via feature normalization,” in *Proceedings of the 40th International Conference on Machine Learning*, 2025.
- 2 **M. Müller**, T. Vlaar, D. Rolnick, and M. Hein, “Normalization layers are all that sharpness-aware minimization needs,” in *Advances in Neural Information Processing Systems*, 2023.
- 3 J. Bitterwolf*, **M. Müller***, and M. Hein, “In or out? fixing imagenet out-of-distribution detection evaluation,” in *Proceedings of the 40th International Conference on Machine Learning*, 2023.*equal contribution.

- 4 M. Andriushchenko, F. Croce, **M. Müller**, M. Hein, and N. Flammarion, “A modern look at the relation between sharpness and generalization,” in *Proceedings of the 40th International Conference on Machine Learning*, 2023.
- 5 **M. Müller** and M. Hein, “Logex: Improved tail detection of extremely rare histopathology classes via guided diffusion,” in *Medical Image Computing and Computer Assisted Intervention - ADSMI Workshop (Oral)*, 2024.
- 6 **M. Müller** and M. Hein, “How to train your vit for ood detection,” in *International Conference on Learning Representations - Workshop on Reliable and Responsible Foundation Models*, 2024.

Books and Chapters

- 1 N. Thuerey, P. Holl, **M. Müller**, P. Schnell, F. Trost, and K. Um, *Physics-based Deep Learning*. WWW, 2021.  URL: <https://physicsbaseddeeplearning.org>.

Journals

- 1 E. Poli, **M. Müller**, H. Zohm, and M. Kovari, “Fast evaluation of the current driven by electron cyclotron waves for reactor studies,” *Physics of Plasmas*, 2018.

Skills

- Languages  German (native), fluent in English and Spanish
- Coding  Python (esp. Pytorch, TensorFlow, sklearn), R, SQL, $\text{\LaTeX}$, AWS, ...

Miscellaneous

Awards and Achievements

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| 2021-now |  ELLIS PhD student , Network for excellent young ML researchers in Europe |
| 2020 |  Master's Thesis selected for BGSE Voice Blog |
| 2013-2021 |  Max Weber Program , Scholarship for gifted students in Bavaria |
| 2016 |  PROSA LMU scholarship for exchange semester in Colombia |
| 2013 |  TUM award for seminar thesis , Visitor jury price for the best poster at students conference |
| |  Awards for outstanding achievements in Maths and Physics , by German Physical Society and German Mathematical Society |

Teaching

-  **Statistical Machine Learning**, TA
-  **Mathematics for Physics**, TA
-  **Experimental Physics for medical students**, lab supervisor

Community Service

-  Frequent reviewer for NeurIPS, ICML and ICLR